

# Design-Weighted LCA: Segments, Prediction, and Handoff

Institutional trust in Mexico: LAPOP AmericasBarometer 2023

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2026-07-21

## 1 Purpose

This script is the fast, interpretive half of the pipeline. It reads the fitted objects the modeling script saved to `{cfg$out_dir}`, drafts (or reads) the segment labels, presents the measurement model with design-based intervals alongside the unweighted poLCA benchmark and its validation verdict, predicts segment membership back to every respondent who answered enough items, profiles the segments on the configured demographics (one page each), and writes the analyst handoff: a labelled `.sav` plus CSVs. Throughout, a latent class is called a **segment**, a word-choice preference; the statistical object is unchanged.

## 2 What the segment labeler is asked

Before any label is trusted, here is the instrument verbatim: the system prompt (the persona) and one real user prompt, built for segment 1 from this survey's own wording and probabilities. Three properties to note while reading: the prompt carries only ONE segment (isolated calls, an empirical design: joint prompts confused near-neighbor segments); every probability is shown against the item's actual response labels; and rule 1 fences the survey context to referent resolution, so nothing outside the shown probabilities may be attributed to a segment. This instrument, including the segment terminology, is certified by `llm_label_obedience_experiment.R`; any edit to it obliges a re-run.

```
meta_preview <- item_meta(survey_dat, cfg$items, cfg$na_codes, questions)
cat("===== SYSTEM PROMPT (persona) =====\n")
```

```
===== SYSTEM PROMPT (persona) =====
```

```
cat(lca_persona(), "\n\n")
```

You are a senior survey methodologist who reads latent class analysis (LCA) measurement models

```
cat("===== USER PROMPT (segment 1) =====\n")
```

```
===== USER PROMPT (segment 1) =====
```

```
cat(prompt_class_label(fit_w, 1, meta_preview, cfg$items, cfg$survey_context), "\n")
```

## SURVEY CONTEXT

These items come from the 2023 AmericasBarometer survey of Mexico, conducted by the LAPOP Lab and the Center for Global Policy Studies.

The battery analyzed here measures trust in national institutions: respondents rate, for each item, the level of trust in the institution on a 6-point scale (1 = "Not at all", 6 = "A great deal").

ONE SEGMENT FROM A LATENT CLASS ANALYSIS (LCA) MEASUREMENT MODEL

SEGMENT 1 (estimated prevalence 33%):

justice\_system "Confianza en el sistema de justicia"

P(Nada)=0.00, P(2)=0.04, P(3)=0.15, P(4)=0.46, P(5)=0.29, P(6)=0.05, P(Mucho)=0.02

electoral\_tribunal "Confianza en el Tribunal Supremo Electoral"

P(Nada)=0.03, P(2)=0.05, P(3)=0.12, P(4)=0.27, P(5)=0.29, P(6)=0.17, P(Mucho)=0.08

armed\_forces "Confianza en las fuerzas armadas"

P(Nada)=0.02, P(2)=0.02, P(3)=0.07, P(4)=0.16, P(5)=0.28, P(6)=0.24, P(Mucho)=0.20

legislature "Confianza en la legislatura"

P(Nada)=0.02, P(2)=0.03, P(3)=0.10, P(4)=0.38, P(5)=0.30, P(6)=0.13, P(Mucho)=0.04

public\_ministry "Confianza en el Ministerio Público"

P(Nada)=0.01, P(2)=0.02, P(3)=0.10, P(4)=0.37, P(5)=0.35, P(6)=0.11, P(Mucho)=0.05

police "Confianza en la policía nacional"

P(Nada)=0.08, P(2)=0.17, P(3)=0.29, P(4)=0.26, P(5)=0.14, P(6)=0.04, P(Mucho)=0.02

auditor "Confianza en la Auditoría Superior de la Federación"

P(Nada)=0.01, P(2)=0.02, P(3)=0.13, P(4)=0.41, P(5)=0.30, P(6)=0.10, P(Mucho)=0.02

political\_parties "Confianza en los partidos políticos"

P(Nada)=0.10, P(2)=0.19, P(3)=0.28, P(4)=0.30, P(5)=0.10, P(6)=0.03, P(Mucho)=0.00

president "Confianza en el presidente"

P(Nada)=0.04, P(2)=0.06, P(3)=0.07, P(4)=0.17, P(5)=0.19, P(6)=0.21, P(Mucho)=0.26

supreme\_court "Confianza en el tribunal supremo"

P(Nada)=0.00, P(2)=0.04, P(3)=0.11, P(4)=0.31, P(5)=0.37, P(6)=0.13, P(Mucho)=0.04

municipality "Confianza en su municipalidad"

P(Nada)=0.01, P(2)=0.04, P(3)=0.14, P(4)=0.37, P(5)=0.34, P(6)=0.08, P(Mucho)=0.02

media "Confianza en los medios de comunicación"

P(Nada)=0.05, P(2)=0.10, P(3)=0.21, P(4)=0.27, P(5)=0.25, P(6)=0.08, P(Mucho)=0.04

elections "Confianza en las elecciones"

P(Nada)=0.02, P(2)=0.05, P(3)=0.21, P(4)=0.30, P(5)=0.26, P(6)=0.13, P(Mucho)=0.02

## TASK

Read this single segment and return: a short DRAFT label (2 to 5 words) for an analyst to refine the survey.

## RULES:

1. Use only the response probabilities and item wording shown. The survey context only clarifies what the items refer to; attribute nothing to the segment that the probabilities do not show.

2. Anchor every statement to the high-probability answers of this segment.
  3. If the profile is diffuse (no clear high-probability answers), say so.
  4. Return only valid JSON: no prose before or after, no markdown fences.
- JSON (one object): {"label": "...", "description": "..."}

### 3 Naming the segments

One rule: `segment_labels.csv` in `{cfg$out_dir}` is used when it exists (validated against K); otherwise the LLM drafts once, one call per segment, and writes it, with any harmonizer edit recorded in `Label_draft`. **Editing that file is taking over the naming**; the labels below are drafts to verify against the profile heatmap.

```
# get_class_labels() implements the one rule above (survey_lca_source.R,
# Section 4). The cache file freezes the LLM output for reproducibility.
# cfg$data, NOT dat_prepared: recode_consecutive() strips attributes, so a
# recoded copy would silently cost the prompts all wording and labels.
segment_labels <- get_class_labels(fit_w, survey_dat, cfg$items, cfg,
                                   questions = set_names(dictionary$question,
                                                         dictionary$item))

knitr::kable(segment_labels,
              caption = "Segment labels and descriptions in use (segment_labels.csv in cfg$out_dir)

```

Table 1: Segment labels and descriptions in use (`segment_labels.csv` in `cfg$out_dir`). Verify each against its response profile before quoting; edit the file to take over naming.

| K | Label                              | Description                                                                                                                                                                                                                                                                                                                                                                  |
|---|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Moderate Institutional Trust       | This segment shows high trust in the president, armed forces, and supreme court, while maintaining moderate trust (concentrated at 4 and 5) across most other national institutions. Trust is notably lower for political parties and the national police.                                                                                                                   |
| 2 | Generalized Institutional Distrust | This segment is characterized by very high probabilities of ‘Nada’ (not at all) across almost all institutions, most notably for political parties (0.85) and the national police (0.76). The only relative exceptions are the president and armed forces, which show more distributed responses and higher probabilities for positive trust levels compared to other items. |
| 3 | Presidential and Military Trust    | This segment is characterized by high trust in the president and armed forces, and contrasted with strong distrust of political parties and the national police.                                                                                                                                                                                                             |
| 4 | Broad High Institutional Trust     | This segment is characterized by very high trust across nearly all institutions, with the highest probabilities of ‘Mucho’ (a lot) for the president (0.82), public ministry (0.74), and armed forces (0.72). Trust is slightly more distributed for political parties and the media, though still leaning toward higher trust levels.                                       |

| K | Label                                 | Description                                                                                                                                                                                                     |
|---|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Concentrated High Institutional Trust | This segment expresses high levels of trust across all institutions, with the strongest probabilities concentrated on the 5, 6, and 7 ratings, particularly for the president, armed forces, and supreme court. |

```
# Display labels for every downstream table and figure: short, wrapped, fixed
# class order so factor levels are stable everywhere.
# Three display tiers, because one label format cannot serve every surface:
#   lab_of      wrapped factor, for FIGURE legends and facets (newlines are fine there)
#   lab_line    single line "Ck: Label", for SMALL tables (a newline inside a kable
#               cell breaks Typst layout)
#   lab_short   "Ck" only, for LARGE tables; the class-label table above is the legend
lab_lev  <- stringr::str_wrap(paste0("S", segment_labels$K, ": ", segment_labels$Label), 22)
lab_of   <- function(k) factor(lab_lev[k], levels = lab_lev)
lab_line <- function(k) paste0("S", k, ": ", segment_labels$Label[k])
lab_short <- function(k) paste0("S", k)
```

## 4 The measurement model: inclusion probabilities, benchmark, and validation

One table carries the design-weighted segment prevalences with their JK<sub>n</sub> intervals, the unweighted poLCA prevalences, and the gap. The printed validation line is the trust check: at unit weights our pseudo-EM and poLCA maximize the same likelihood, so their prevalences must match to optimization tolerance before the weighted output deserves belief.

```
# The verdict is READ from validation.parquet, frozen by the modeling script,
# so both documents print the same criterion on the same numbers: max of the
# prevalence gap and the conditional-probability gap, threshold 0.005.
cat(sprintf(paste0("VALIDATION (unit weights, pseudo-EM vs poLCA): ",
  "max |pi gap| = %.5f, max |rho gap| = %.5f -> %s\n"),
  validation$max_pi_gap, validation$max_rho_gap,
  ifelse(validation$match, "MATCH (implementations agree)",
    "MISMATCH - investigate before trusting weighted results")))
```

```
VALIDATION (unit weights, pseudo-EM vs poLCA): max |pi gap| = 0.00000, max |rho gap| = 0.00000
```

```
# ci_tbl columns (from the modeling script): kind, item, category, class,
# estimate, se, lo, hi. Renamed here to short names for the display code.
pi_ci <- ci_tbl |> dplyr::filter(kind == "pi") |>
  transmute(K = class, est = estimate, lo, hi)
incl_tbl <- pi_ci |>
```

```

mutate(Segment = lab_line(K),
       `Weighted pi [95% CI]` = sprintf("%.3f [%.3f, %.3f]", est, lo, hi),
       `poLCA (unweighted)` = sprintf("%.3f", pl_aligned$pi[K]),
       Gap = sprintf("%+.3f", est - pl_aligned$pi[K])) |>
dplyr::select(Segment:Gap)
knitr::kable(incl_tbl, align = "lrrrr",
             caption = "Segment inclusion probabilities: design-weighted estimates with JKn intervals, the unweighted poLCA benchmark, and the weighting gap. The validation line above certifies the implementations agree at unit weights.")

```

Table 2: Segment inclusion probabilities: design-weighted estimates with JK<sub>n</sub> intervals, the unweighted poLCA benchmark, and the weighting gap. The validation line above certifies the implementations agree at unit weights.

| Segment                                   | Weighted pi [95% CI] | poLCA<br>(unweighted) | Gap    |
|-------------------------------------------|----------------------|-----------------------|--------|
| S1: Moderate Institutional Trust          | 0.331 [0.293, 0.372] | 0.336                 | -0.005 |
| S2: Generalized Institutional Distrust    | 0.107 [0.086, 0.132] | 0.107                 | +0.000 |
| S3: Presidential and Military Trust       | 0.242 [0.204, 0.285] | 0.239                 | +0.004 |
| S4: Broad High Institutional Trust        | 0.111 [0.093, 0.134] | 0.111                 | -0.000 |
| S5: Concentrated High Institutional Trust | 0.208 [0.173, 0.249] | 0.207                 | +0.001 |

```

pi_ci |>
mutate(Segment = lab_of(K)) |>
ggplot(aes(Segment, est)) +
geom_pointrange(aes(ymin = lo, ymax = hi), linewidth = 0.5) +
geom_point(aes(y = pl_aligned$pi[K]), shape = 5, size = 2.6) +
scale_y_continuous(limits = c(0, NA)) +
labs(x = NULL, y = "Inclusion probability") +
theme_lca() +
theme(axis.text.x = element_text(angle = 20, hjust = 1))

```

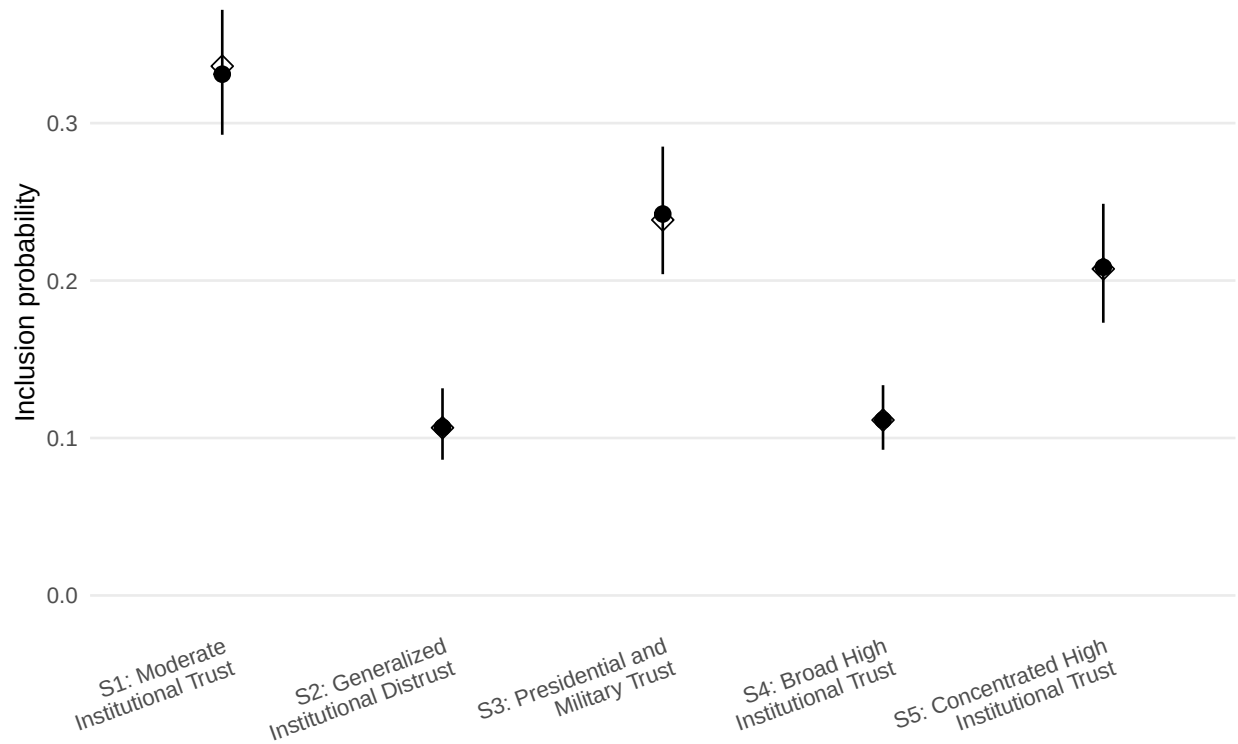


Figure 1: Weighted segment prevalences with 95% JKn intervals (points and bars) against the unweighted poLCA benchmark (open diamonds). Gaps are what the design corrects.

## 5 Segment profiles: conditional response probabilities

```
ci_tbl |>
  dplyr::filter(kind == "rho") |>
  mutate(Segment = lab_of(class),
         item = factor(item, levels = cfg$items),
         est = estimate) |>
  ggplot(aes(factor(category), item, fill = est)) +
  geom_tile() +
  geom_text(aes(label = sprintf("%.2f", est)), size = 2.1) +
  scale_fill_viridis_c(name = "P", limits = c(0, 1)) +
  facet_wrap(~ Segment, ncol = min(3, K_star)) +
  labs(x = "Response category", y = NULL) +
  theme_lca() +
  theme(axis.text.y = element_text(size = 7))
```

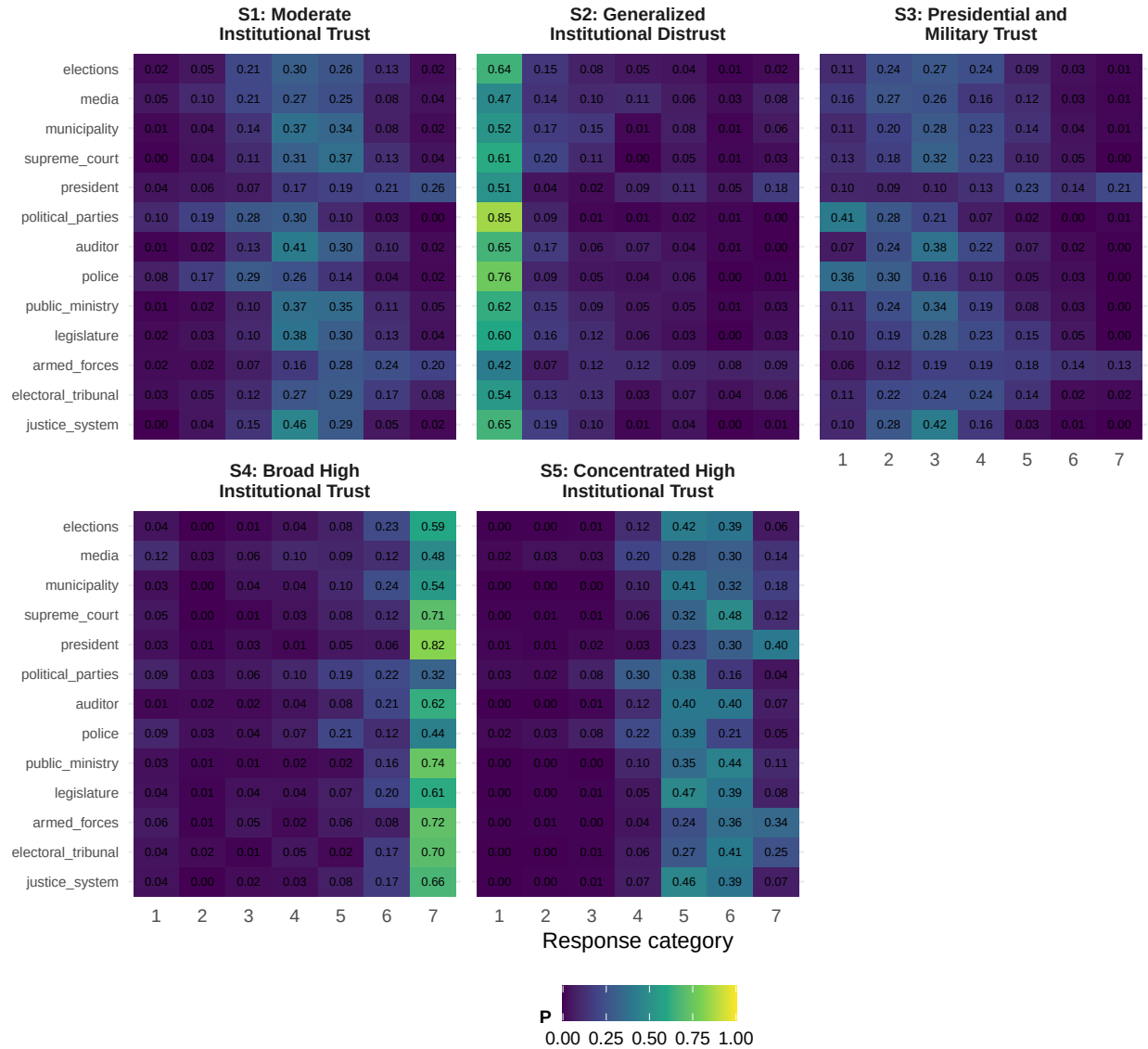


Figure 2:  $P(\text{response} \mid \text{segment})$  for every item. Panel headers carry the drafted segment labels; verify each label against its panel before quoting it.

## 6 Predicting segment membership

Prediction runs on the FULL preserved frame: local independence lets a missing item drop out of the within-segment product, so partial responders are predictable, with `cfg$min_items_predict` as the evidence floor (below it, no prediction rather than a guess). Items must be coded exactly as fitted; the config script guarantees that here.

```
pred <- predict_segments(survey_dat_original, fit_w, cfg$items,
                        min_items = cfg$min_items_predict)
survey_scored <- bind_cols(survey_dat_original, pred)
```

```
cat(sprintf(
  "Predicted %d complete and %d incomplete cases of %d total (floor: %d items); mean max poster
  sum(!is.na(pred$segment) & survey_scored$keep),
  sum(!is.na(pred$segment) & !survey_scored$keep),
  nrow(survey_scored), cfg$min_items_predict,
  mean(pred$max_posterior, na.rm = TRUE)))
```

Predicted 1430 complete and 188 incomplete cases of 1622 total (floor: 7 items); mean max poster

## 7 Segment composition by demographics

BCH-corrected segment prevalence within each demographic level (Bolck, Croon & Hagenaars 2004; Vermunt 2010): modal assignments are reweighted by the inverse of the design-weighted classification-error matrix, so the cross-tabs are unbiased for true-segment composition rather than diluted by classification error. Intervals are JK<sub>n</sub> on the logit scale with the design  $t$  critical value; cells outside (0, 1), possible because BCH weights can be negative, fall back to truncated Wald and are footnoted. Step-1 parameter uncertainty is not propagated, the standard treatment; with entropy 0.907 the understatement is minor (Bakk, Oberski & Vermunt 2014). One demographic per page, table above plot, so nothing crowds.

```
# The segment-membership target is the soft posterior attached in the setup
# chunk; post_cols are the K posterior columns in segment order. des and rep_des
# (the design and its JKn replicates) were also built in setup.
post_cols <- paste0("post_class", seq_len(K_star))

# Fixed layout for the flattened statistic: every (demographic, level, class) cell,
# levels in factor order, class varying fastest within a level. dom_meta names and
# orders the estimate vector so nothing is recovered by string parsing later.
dom_meta <- map_dfr(cfg$aux, function(v)
  expand_grid(plevel = levels(dat_prepared[[v]]), class = seq_len(K_star)) |>
  mutate(pvar = v) |>
  dplyr::select(pvar, plevel, class) |>
  mutate(idx = row_number())

# The statistic survey::withReplicates evaluates on each replicate: the BCH
# three-step correction. post and modal are FIXED at the full-sample
# measurement model (step-1 uncertainty not propagated, standard practice);
# the classification-error matrix D and the correction weights u are rebuilt
# inside every replicate from the replicate weights, so the intervals carry
# the design variance of the classification table and the cross-tab jointly.
# For each demographic: a level-indicator matrix, then the weighted mean of
# each BCH-corrected membership column within each level,
# crossprod(M, w * U) / crossprod(M, w), flattened to ONE numeric vector in
# dom_meta's order (level-major, class-minor) for the replicate covariance.
domain_theta <- function(w_rep, data) {
```



```

P <- as.matrix(data[, post_cols, drop = FALSE])      # n x K, fixed across replicates
U <- bch_weights(P, max.col(P, ties.method = "first"), w_rep) # n x K BCH
wU <- w_rep * U                                     # n x K, weighted corrections
flat <- map(cfg$aux, function(v) {
  f <- factor(data[[v]], levels = levels(dat_prepared[[v]]))
  M <- model.matrix(~ f - 1)                        # n x L, one column per level
  num <- crossprod(M, wU)                          # L x K: corrected sums per level
  den <- as.numeric(crossprod(M, w_rep))            # L: weighted total per level
  as.vector(t(num / den))                          # level-major, class-minor
})
unlist(flat, use.names = FALSE)
}

dom_rep <- withReplicates(rep_des, domain_theta)      # design-based covariance of every cell
dom_est <- domain_theta(weights(des), dat_prepared)  # full-sample point estimates
dom_V <- vcov(dom_rep)                             # full covariance, for the contrasts b
dom_se <- sqrt(diag(dom_V))
stopifnot(length(dom_est) == nrow(dom_meta))

# Per-level BCH-corrected prevalence with a design-based interval: logit-scale
# delta method (se_logit = se / (p (1 - p))), back-transformed, with the design
# t critical value. BCH point estimates can fall outside (0, 1); the logit
# transform is undefined there, so those cells fall back to Wald truncated to
# [0, 1] and are flagged in `wald_fallback` for the table footnote.
crit <- qt(0.975, df = survey::degf(rep_des))
dom_tbl <- dom_meta |>
  mutate(p = as.numeric(dom_est), se = as.numeric(dom_se)) |>
  mutate(inside = p > 0 & p < 1,
         p_c = pmin(pmax(p, 1e-6), 1 - 1e-6),
         gse = se / (p_c * (1 - p_c)),
         lo = if_else(inside, plogis(qlogis(p_c) - crit * gse),
                      pmax(0, p - crit * se)),
         hi = if_else(inside, plogis(qlogis(p_c) + crit * gse),
                      pmin(1, p + crit * se)),
         wald_fallback = !inside) |>
  dplyr::select(pvar, plevel, class, p, se, lo, hi, wald_fallback)

walk(cfg$aux, function(v) {
  d <- dom_tbl |> dplyr::filter(pvar == v)
  d |>
    transmute(Level = plevel, Segment = lab_short(class),
              cell = sprintf("%.2f [%.2f, %.2f]", p, lo, hi)) |>
    pivot_wider(names_from = Segment, values_from = cell) |>
    # format = "pipe" + cat(): guaranteed emission under results: asis, and a
    # Pandoc table that wraps instead of overflowing (the kable |> print()
    # pattern is silently swallowed by some render paths).
    knitr::kable(format = "pipe", align = str_c("l", strrep("r", K_star)),

```

```

        caption = paste0("BCH-corrected segment prevalence within ", v,
                          " (95% JKn intervals, logit scale; rows sum to one across segments; S1..S5",
                          K_star, " named in the segment-label table",
                          if (any(d$wald_fallback))
                            "; cells outside (0, 1) use truncated Wald intervals" else
                            ").")) |>

    cat(sep = "\n")
    cat("\n\n")
    p <- d |>
    mutate(Segment = lab_of(class)) |>
    ggplot(aes(p, plevel, color = Segment)) +
    geom_pointrange(aes(xmin = lo, xmax = hi), size = 0.35, linewidth = 0.4,
                   position = position_dodge(width = 0.6)) +
    scale_color_viridis_d(begin = 0.1, end = 0.8) +
    scale_x_continuous(limits = c(0, NA)) +
    labs(x = "Prevalence of segment membership", y = NULL, color = NULL,
         title = v) +
    theme_lca()
    print(p)
    cat("\n\n\\newpage\n\n")
  })

```

Table 3: BCH-corrected segment prevalence within age\_cat (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level | S1   |              | S2   |              | S3   |              | S4   |              | S5                |
|-------|------|--------------|------|--------------|------|--------------|------|--------------|-------------------|
| 18-29 | 0.35 | [0.31, 0.41] | 0.09 | [0.07, 0.12] | 0.25 | [0.22, 0.29] | 0.09 | [0.07, 0.12] | 0.21 [0.18, 0.25] |
| 30-44 | 0.32 | [0.27, 0.38] | 0.10 | [0.08, 0.14] | 0.26 | [0.21, 0.32] | 0.10 | [0.07, 0.13] | 0.22 [0.17, 0.27] |
| 45-59 | 0.32 | [0.27, 0.38] | 0.12 | [0.09, 0.17] | 0.26 | [0.21, 0.31] | 0.13 | [0.10, 0.17] | 0.17 [0.13, 0.22] |
| 60+   | 0.31 | [0.25, 0.39] | 0.11 | [0.08, 0.16] | 0.17 | [0.12, 0.24] | 0.15 | [0.11, 0.20] | 0.26 [0.20, 0.32] |

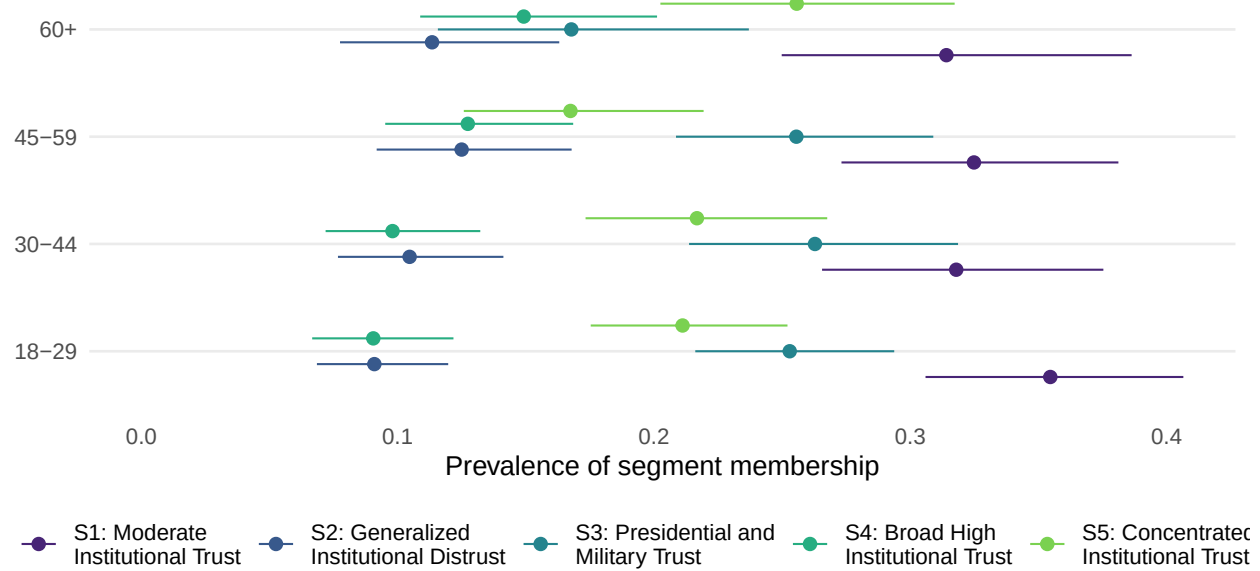


Table 4: BCH-corrected segment prevalence within male (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level  | S1                | S2                | S3                | S4                | S5                |
|--------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Female | 0.35 [0.31, 0.38] | 0.12 [0.10, 0.14] | 0.24 [0.20, 0.27] | 0.11 [0.09, 0.14] | 0.19 [0.16, 0.22] |
| Male   | 0.31 [0.28, 0.35] | 0.10 [0.08, 0.12] | 0.25 [0.21, 0.29] | 0.11 [0.09, 0.14] | 0.23 [0.20, 0.27] |

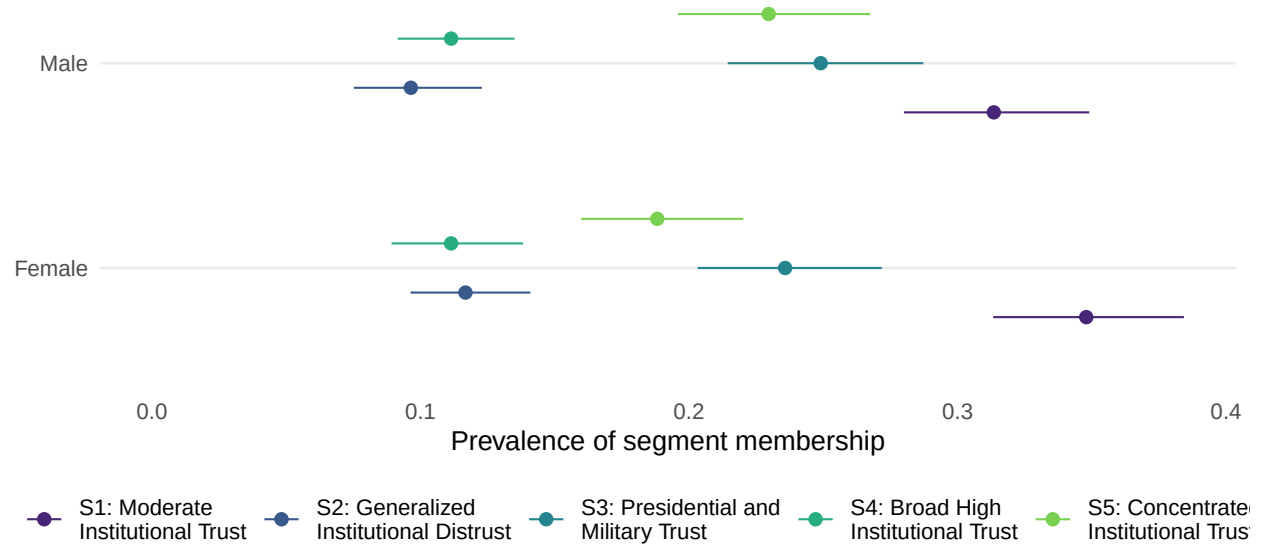


Table 5: BCH-corrected segment prevalence within Edu (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level     | S1                | S2                | S3                | S4                | S5                |
|-----------|-------------------|-------------------|-------------------|-------------------|-------------------|
| higher_ed | 0.31 [0.25, 0.37] | 0.10 [0.07, 0.14] | 0.36 [0.30, 0.43] | 0.07 [0.04, 0.11] | 0.16 [0.12, 0.22] |
| none      | 0.11 [0.03, 0.36] | 0.21 [0.08, 0.43] | 0.12 [0.03, 0.36] | 0.41 [0.22, 0.64] | 0.15 [0.05, 0.38] |
| primary   | 0.35 [0.29, 0.42] | 0.10 [0.06, 0.14] | 0.17 [0.13, 0.23] | 0.17 [0.13, 0.22] | 0.21 [0.15, 0.28] |
| secondary | 0.34 [0.31, 0.37] | 0.11 [0.09, 0.13] | 0.23 [0.20, 0.26] | 0.10 [0.08, 0.12] | 0.23 [0.20, 0.26] |

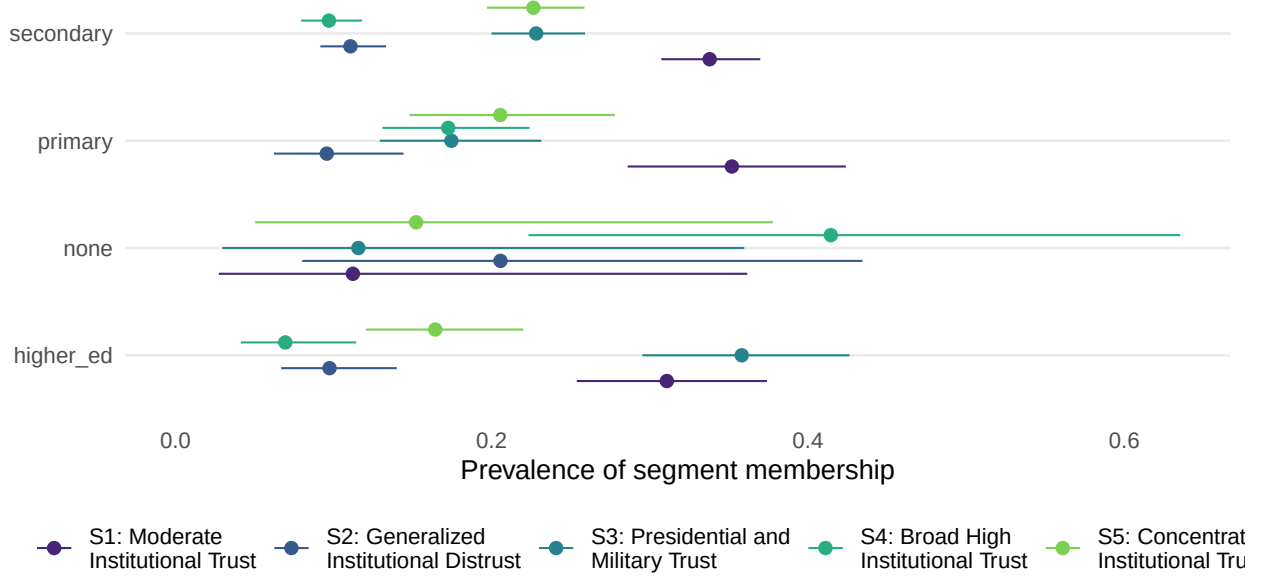


Table 6: BCH-corrected segment prevalence within Urban (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level | S1                | S2                | S3                | S4                | S5                |
|-------|-------------------|-------------------|-------------------|-------------------|-------------------|
| rural | 0.35 [0.29, 0.41] | 0.09 [0.06, 0.14] | 0.18 [0.14, 0.22] | 0.12 [0.09, 0.17] | 0.25 [0.19, 0.33] |
| urban | 0.33 [0.30, 0.36] | 0.11 [0.09, 0.13] | 0.26 [0.23, 0.29] | 0.11 [0.09, 0.13] | 0.20 [0.17, 0.23] |

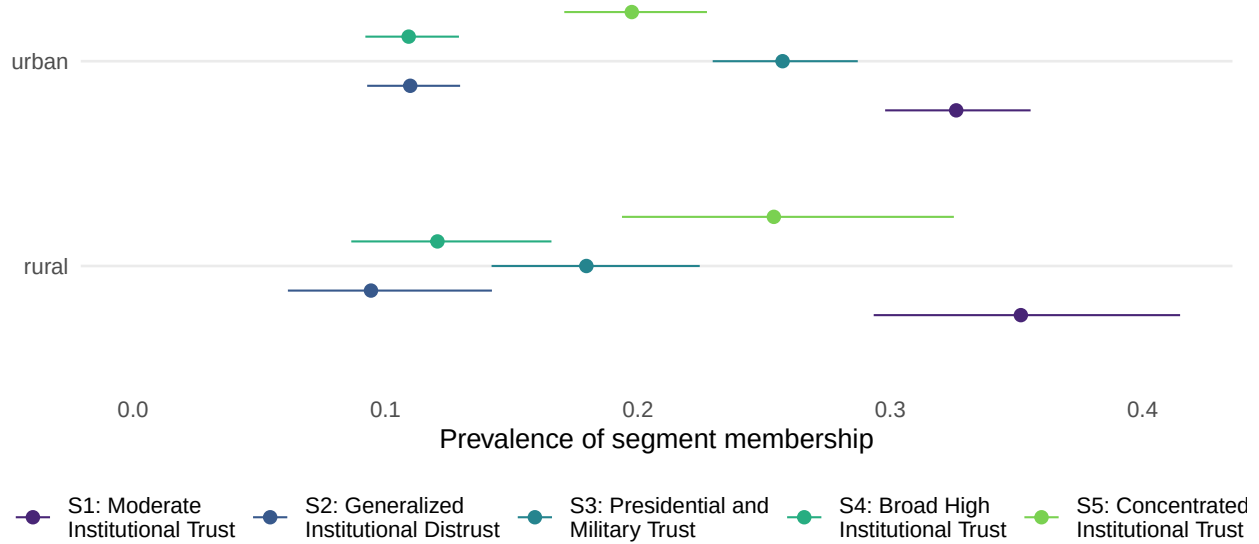


Table 7: BCH-corrected segment prevalence within Employment (95% JK<sub>kn</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level        | S1                | S2                | S3                | S4                | S5                |
|--------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Employed     | 0.33 [0.30, 0.37] | 0.12 [0.10, 0.15] | 0.24 [0.21, 0.28] | 0.10 [0.08, 0.12] | 0.20 [0.17, 0.24] |
| house_keeper | 0.31 [0.26, 0.36] | 0.09 [0.07, 0.13] | 0.25 [0.20, 0.30] | 0.13 [0.10, 0.17] | 0.22 [0.18, 0.27] |
| Retired      | 0.36 [0.26, 0.48] | 0.10 [0.05, 0.19] | 0.19 [0.12, 0.29] | 0.18 [0.11, 0.28] | 0.17 [0.10, 0.27] |
| Student      | 0.40 [0.30, 0.52] | 0.07 [0.03, 0.17] | 0.19 [0.12, 0.29] | 0.09 [0.05, 0.17] | 0.25 [0.17, 0.36] |
| Unemployed   | 0.33 [0.23, 0.44] | 0.09 [0.05, 0.16] | 0.31 [0.22, 0.42] | 0.09 [0.05, 0.16] | 0.19 [0.11, 0.29] |

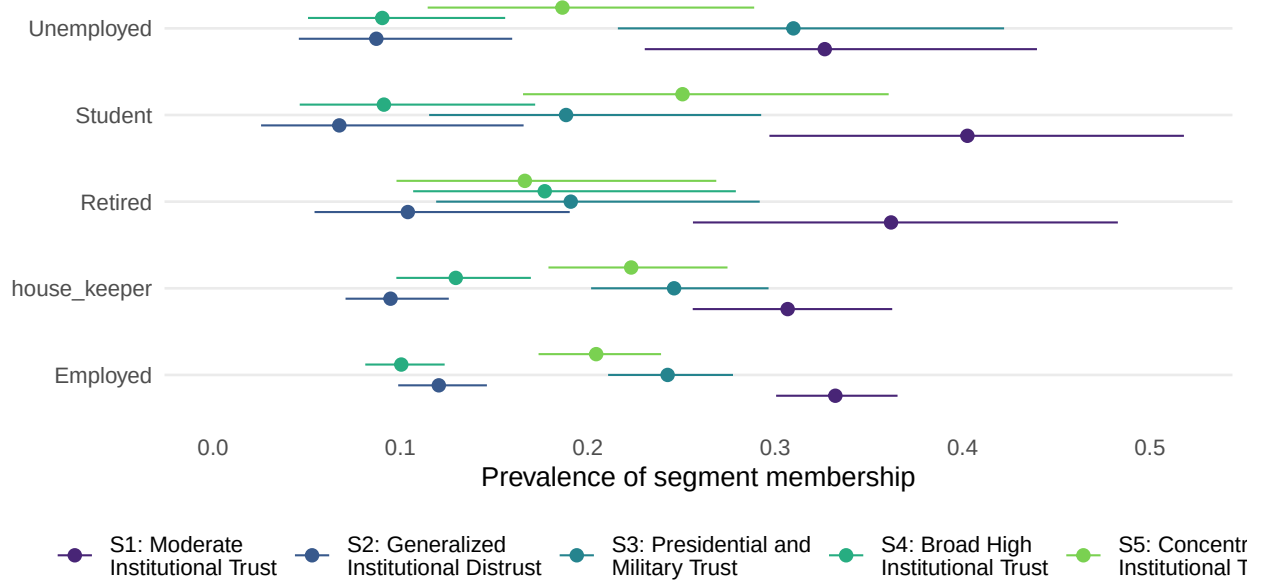


Table 8: BCH-corrected segment prevalence within satis\_demo (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table).

| Level             | S1                | S2                | S3                | S4                | S5                |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Dissatisfied      | 0.34 [0.30, 0.39] | 0.15 [0.12, 0.19] | 0.33 [0.29, 0.38] | 0.04 [0.02, 0.06] | 0.14 [0.10, 0.18] |
| Satisfied         | 0.36 [0.32, 0.39] | 0.05 [0.04, 0.07] | 0.19 [0.16, 0.22] | 0.14 [0.11, 0.17] | 0.26 [0.23, 0.30] |
| Very dissatisfied | 0.22 [0.13, 0.34] | 0.34 [0.25, 0.45] | 0.28 [0.19, 0.40] | 0.05 [0.02, 0.11] | 0.11 [0.05, 0.21] |
| Very satisfied    | 0.18 [0.11, 0.29] | 0.06 [0.02, 0.13] | 0.19 [0.11, 0.30] | 0.34 [0.25, 0.44] | 0.24 [0.15, 0.35] |

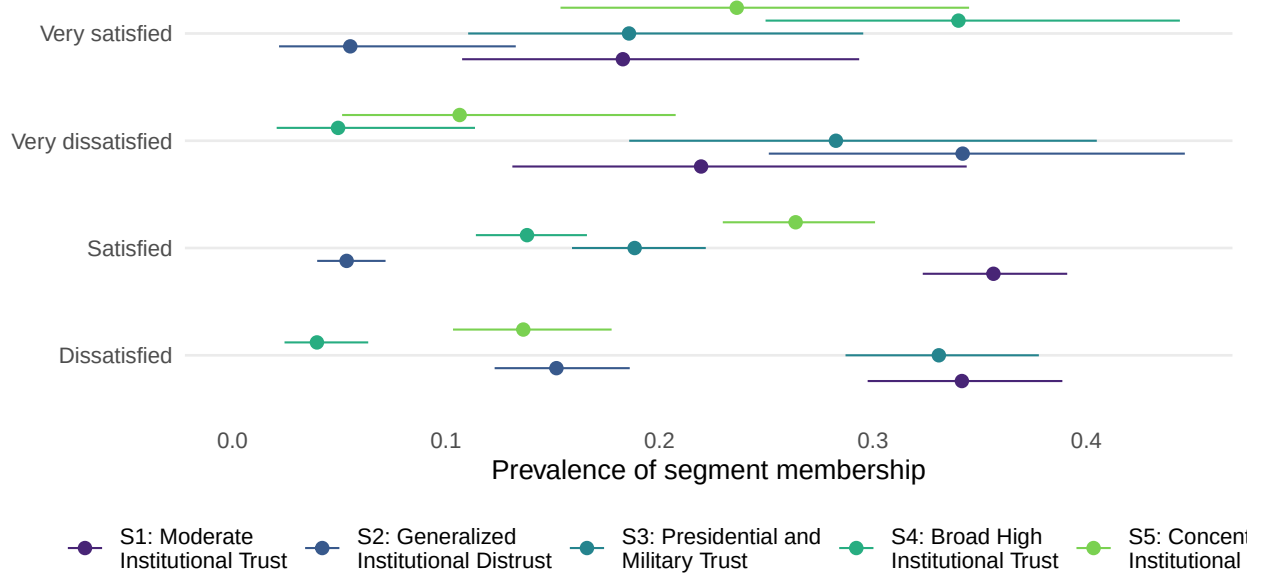
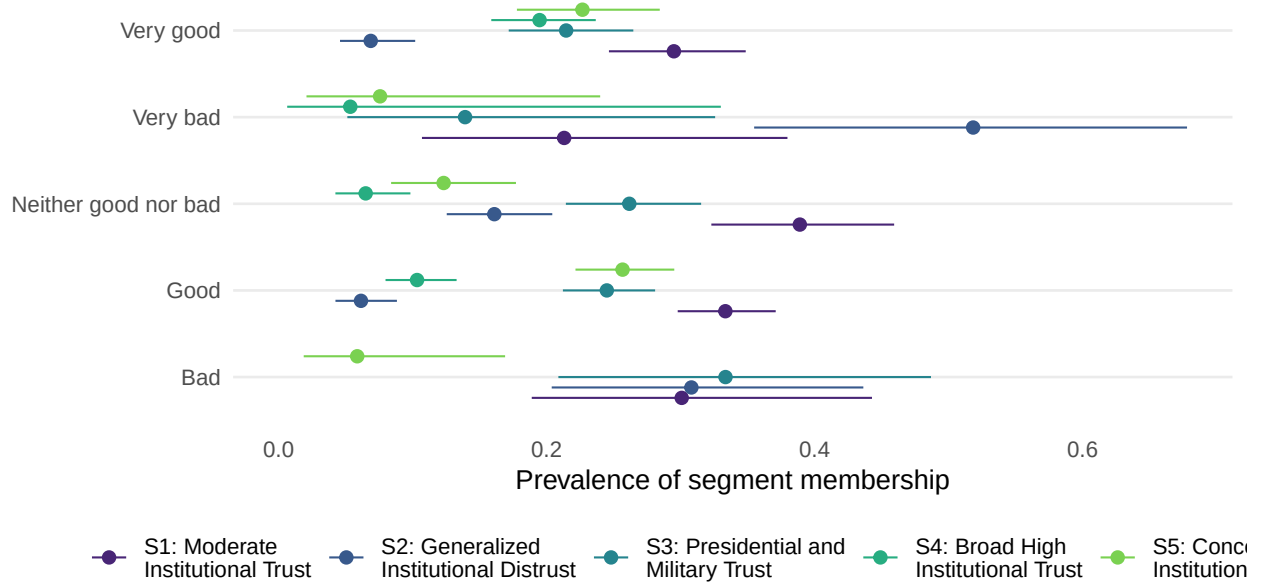




Table 9: BCH-corrected segment prevalence within prez\_rating (95% JK<sub>n</sub> intervals, logit scale; rows sum to one across segments; S1..S5 named in the segment-label table; cells outside (0, 1) use truncated Wald intervals).

| Level                | S1                | S2                | S3                | S4                  | S5                |
|----------------------|-------------------|-------------------|-------------------|---------------------|-------------------|
| Bad                  | 0.30 [0.19, 0.44] | 0.31 [0.20, 0.44] | 0.33 [0.21, 0.49] | -0.00 [0.00, -0.00] | 0.06 [0.02, 0.17] |
| Good                 | 0.33 [0.30, 0.37] | 0.06 [0.04, 0.09] | 0.24 [0.21, 0.28] | 0.10 [0.08, 0.13]   | 0.26 [0.22, 0.30] |
| Neither good nor bad | 0.39 [0.32, 0.46] | 0.16 [0.13, 0.20] | 0.26 [0.21, 0.32] | 0.07 [0.04, 0.10]   | 0.12 [0.08, 0.18] |
| Very bad             | 0.21 [0.11, 0.38] | 0.52 [0.35, 0.68] | 0.14 [0.05, 0.33] | 0.05 [0.01, 0.33]   | 0.08 [0.02, 0.24] |
| Very good            | 0.30 [0.25, 0.35] | 0.07 [0.05, 0.10] | 0.21 [0.17, 0.26] | 0.19 [0.16, 0.24]   | 0.23 [0.18, 0.28] |



## 8 The handoff

Three artifacts: the labelled `.sav` for analysts (numeric `segment` with the LLM/analyst labels as value labels, NA below the prediction floor, plus `segment_label` as text and the diagnostics from prediction), and CSVs of the measurement model and the demographic results.

```
seg_labelled <- haven::labelled(  
  survey_scored$segment,  
  labels = set_names(segment_labels$K, segment_labels$Label),  
  label = "Segment (design-weighted LCA)")  
out_sav <- survey_scored |>  
  mutate(segment = seg_labelled,  
         segment_label = ifelse(is.na(segment), NA_character_,  
                                segment_labels$Label[survey_scored$segment]))  
haven::write_sav(out_sav, file.path(cfg$out_dir, "survey_with_segments.sav"))  
  
readr::write_csv(ci_tbl |>  
  mutate(segment_label = segment_labels$Label[class]),  
  file.path(cfg$out_dir, "measurement_model_with_CIs.csv"))  
readr::write_csv(dom_tbl |>  
  transmute(demographic = pvar, level = plevel,  
            segment = class,  
            segment_label = segment_labels$Label[class],  
            prevalence = p, lo, hi),  
  file.path(cfg$out_dir, "segment_demographics.csv"))  
cat("Wrote survey_with_segments.sav (", sum(!is.na(out_sav$segment)), "assigned ),",  
    "measurement_model_with_CIs.csv, segment_demographics.csv to", cfg$out_dir, "\n")
```

Wrote survey\_with\_segments.sav ( 1618 assigned ), measurement\_model\_with\_CIs.csv, segment\_demo